EVERY TORSOR UNDER THE UNIVERSAL JACOBIAN OVER M_G IS A RELATIVE PICARD COMPONENT

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Allowed standard prerequisites (editorial).

- Franchetta theorem, Teichmuller uniformization, mapping-class homology stability and Leray/Kummer sequences. Original complete reference data and author-local formulations retained.

1. Introduction

Throughout we work over the field of complex numbers $\mathbb{C}$.

1.1. **Main result.** Let $\mathcal{M}_g$ be the moduli stack of smooth genus-g curves, and let $\pi\colon \mathcal{M}_{g,1} \to \mathcal{M}_g$ be the universal family. The relative Jacobian $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ is a family of abelian varieties over $\mathcal{M}_g$. We show that:

Theorem 1. *For $g \geq 3$, we have*

$$\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g, \mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \cong \mathbb{Z}/(2g-2)\mathbb{Z} \cdot [\mathrm{Pic}^1_{\mathcal{M}_{g,1}/\mathcal{M}_g}].$$

This establishes a Franchetta-type result: For $g \geq 3$, all torsors under the universal Jacobian arise from the "obvious" construction.

1.2. **Sketch of the proof.** First, the strong Franchetta theorem [Sch03, Theorem 5.1] reduces the computation in Theorem 1 to proving the vanishing $\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g, \mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g}) = 0$.

Then we show that for $g \geq 3$, the pullback map $\pi^*\colon \mathrm{Br}'(\mathcal{M}_g) \to \mathrm{Br}'(\mathcal{M}_{g,1})$ is surjective. For $g \geq 4$ this is known by [DLP25, Theorem 4.1]. For $g = 3$ we perform an explicit analysis of the Leray spectral sequence, where we use a topological version of the Franchetta conjecture [HR01, Theorem 24], which builds upon results due to Morita. We leave details of the proof to the end.

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Now the Leray spectral sequence for $\pi\colon \mathcal{M}_{g,1} \to \mathcal{M}_g$ further reduces the question to proving the injectivity of the pullback map $\pi^*\colon \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \to \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$.

Finally, via the comparison theorem and the Teichmüller uniformization, this injectivity follows from stability results [Wah13, Theorem 1.2] on the homology of mapping class groups of oriented surfaces.

2. Preparation

2.1. **The strong Franchetta theorem.** For $g \geq 3$, the moduli stack $\mathcal{M}_{g,n}$ of n-pointed smooth genus-g curves has trivial generic stabilizer, hence the generic fiber of the forgetful morphism between coarse moduli spaces $\pi\colon \overline{M}_{g,n+1} \to \overline{M}_{g,n}$ is a smooth genus-g curve.

Proposition 1. [Sch03, Theorem 5.1] *Let k be a field, and $g \geq 3$ and $n \geq 0$ be integers. Let $\eta_n \in \overline{M}_{g,n}$ be the generic point in the moduli space of n-pointed stable curves of genus $g \geq 3$, and $C = \overline{M}_{g,n+1}$ the tautological curve. Then the marked points $c_1, \cdots, c_n \in C_{\eta_n}$ and the canonical class $K_{C_{\eta_n}}$ freely generate the group of η_n- points of* $\mathrm{Pic}_{C_{\eta_n}/\eta_n}$.

2.2. **The Teichmüller Uniformization.** Let $\mathcal{M}^{\text{an}}_{g,n}$ be the analytification of the Deligne–Mumford stack $\mathcal{M}_{g,n}$. The Teichmüller uniformization gives a presentation

$$\mathcal{M}^{\text{an}}_{g,n} \cong [\mathcal{T}_{g,n}/\Gamma^n_g],$$

where $\mathcal{T}_{g,n}$ is homeomorphic to $\mathbb{C}^{3g-3+n}$, see [ACG11, Chapter XV].

As a consequence, the Hochschild-Serre spectral sequence for singular cohomology

$$\mathrm{H}^p(\Gamma^n_g, \mathrm{H}^q(\mathcal{T}_{g,n}, \mathbb{Q}_\ell/\mathbb{Z}_\ell)) \Rightarrow \mathrm{H}^{p+q}(\mathcal{M}^{\text{an}}_{g,n}, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$$

degenerates to one row, giving natural isomorphisms $\mathrm{H}^p(\mathcal{M}^{\text{an}}_{g,n}, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \cong \mathrm{H}^p(\Gamma^n_g, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$.

2.3. **The Brauer group of $\mathcal{M}_{g,n}$.** Since $\mathcal{M}_{g,n}$ are regular and noetherian Deligne–Mumford stacks, the étale cohomology groups $\mathrm{H}^2_{\text{ét}}(\mathcal{M}_{g,n}, \mathbb{G}_m)$ are torsion [AM20, Theorem 2.5(iii)], hence the cohomological Brauer groups $\mathrm{Br}'(\mathcal{M}_{g,n}) := \mathrm{H}^2_{\text{ét}}(\mathcal{M}_{g,n}, \mathbb{G}_m)_{\text{tors}}$ coincide with the étale cohomology groups $\mathrm{H}^2_{\text{ét}}(\mathcal{M}_{g,n}, \mathbb{G}_m)$.

Proposition 2. *Let $g \geq 3$ and $n \geq 0$ be integers, then*

$$\mathrm{Br}'(\mathcal{M}_{g,n}) = \begin{cases} 0 & g \geq 4, n \geq 0 \\ \mathbb{Z}/2\mathbb{Z} & g = 3, n = 0, 1 \end{cases}$$

Moreover, the pullback map $\pi^*\colon \mathrm{Br}'(\mathcal{M}_3) \to \mathrm{Br}'(\mathcal{M}_{3,1})$ *is an isomorphism.*

Proof. For $g \geq 4$, we refer to [FP21, Theorem 4.1]. For $g = 3$, by [DLP25, Theorem 1.1] we know that $\mathrm{Br}'(\mathcal{M}_3) \cong \mathbb{Z}/2\mathbb{Z}$, and by [Lor25, Corollary 3.22], we know that $\pi^*\colon \mathrm{Br}'(\mathcal{M}_3) \to \mathrm{Br}'(\mathcal{M}_{3,n})$ is injective. We aim to show that $\pi^*\colon \mathrm{Br}'(\mathcal{M}_3) \to \mathrm{Br}'(\mathcal{M}_{3,1})$ is surjective. Since both sides are torsion, it suffices to show surjectivity on the m-torsion parts for all $m > 0$. Comparison of Kummer sequence gives us the commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathrm{Pic}(\mathcal{M}_3) \otimes \mathbb{Z}/m\mathbb{Z} & \xrightarrow{\delta} & \mathrm{H}^2_{\text{ét}}(\mathcal{M}_3, \mu_m) & \longrightarrow & \mathrm{Br}'(\mathcal{M}_3)[m] & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle \pi^*} & & \downarrow{\scriptstyle \pi^*} & & \downarrow{\scriptstyle \pi^*} & & \\ 0 & \longrightarrow & \mathrm{Pic}(\mathcal{M}_{3,1}) \otimes \mathbb{Z}/m\mathbb{Z} & \xrightarrow{\delta} & \mathrm{H}^2_{\text{ét}}(\mathcal{M}_{3,1}, \mu_m) & \longrightarrow & \mathrm{Br}'(\mathcal{M}_{3,1})[m] & \longrightarrow & 0 \end{array}$$

By the snake lemma, it suffices to show that

$$\mathrm{H}^2_{\text{ét}}(\mathcal{M}_{3,1}, \mu_m) = \pi^*\mathrm{H}^2_{\text{ét}}(\mathcal{M}_3, \mu_m) + \delta\left(\mathrm{Pic}(\mathcal{M}_{3,1})\right).$$

The Leray spectral sequence for μ_m along $\pi\colon \mathcal{M}_{3,1} \to \mathcal{M}_3$ induces a filtration $F^0 \supseteq F^1 \supseteq F^2$, where $F^0 = \mathrm{H}^2_{\text{ét}}(\mathcal{M}_{3,1}, \mu_m)$ and $F^2 = \pi^*\mathrm{H}^2_{\text{ét}}(\mathcal{M}_3, \mu_m)$. It suffices to show that any class in $\mathrm{H}^2_{\text{ét}}(\mathcal{M}_{3,1}, \mu_m)$ can be modified by $\delta(\mathrm{Pic}(\mathcal{M}_{3,1}))$ so that the image in F^0/F^1 and F^1/F^2 both equal to zero. We leave the details to Proposition 4 in Section 4. □

2.4. **Homology stability theorem.** Let S_g be a surface of genus g, and let $x \in S_g$ be a fixed point. Let $S_g^1 = S_g - \{x\}$ be the punctured surface. Let $S_{g,1}$ be the surface with boundary, obtained from S_g by removing an open disc around x. We have natural inclusions

$$S_{g,1} \subseteq S_g^1 \subseteq S_g.$$

Let $\Gamma_g = \pi_0\mathrm{Diff}^+(S_g)$ be the group of components of the orientation preserving diffeomorphism group of S_g. Let $\Gamma_{g,1} = \pi_0\mathrm{Diff}^+(S_{g,1}; \partial S_{g,1})$ and $\Gamma_g^1 = \pi_0\mathrm{Diff}^+(S_g; \{x\})$ be the group of components of diffeomorphisms that are identity on the boundary of $S_{g,1}$ and on the point $\{x\}$, respectively.

Let $\delta_g\colon \Gamma_{g,1} \to \Gamma_g$ be the map induced by the inclusion $S_{g,1} \hookrightarrow S_g$. Then we have:

Proposition 3. [Wah13, Theorem 1.2] *The map* $\mathrm{H}_k(\delta_g)\colon \mathrm{H}_k(\Gamma_{g,1}, \mathbb{Z}) \to \mathrm{H}_k(\Gamma_g, \mathbb{Z})$ *is surjective for* $k \leq \frac{2}{3}g + 1$ *and an isomorphism for* $k \leq \frac{2}{3}g$.

Let $\epsilon_g\colon \Gamma_g^1 \to \Gamma_g$ be the map induced by the inclusion $S_g^1 \subseteq S_g$. Since $\mathrm{H}_k(\delta_g)$ factors through $\mathrm{H}_k(\epsilon_g)\colon \mathrm{H}_k(\Gamma_g^1,\mathbb{Z}) \to \mathrm{H}_k(\Gamma_g,\mathbb{Z})$, the surjectivity of $\mathrm{H}_k(\delta_g)$ implies the surjectivity of $\mathrm{H}_k(\epsilon_g)$. Therefore we have:

Corollary 1. *The map* $\mathrm{H}_k(\epsilon_g)\colon \mathrm{H}_k(\Gamma_g^1,\mathbb{Z}) \to \mathrm{H}_k(\Gamma_g,\mathbb{Z})$ *is surjective for* $k \leq \frac{2}{3}g+1$.

Taking $k=3$, we see that $\mathrm{H}_3(\epsilon_g)\colon \mathrm{H}_3(\Gamma_g^1,\mathbb{Z}) \twoheadrightarrow \mathrm{H}_3(\Gamma_g,\mathbb{Z})$ for all $g \geq 3$.

3. Proof of Theorem 1

3.1. **Reduction to vanishing.** Consider the short exact sequence of étale sheaves on $\mathcal{M}_g$

$$0 \longrightarrow \mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g} \longrightarrow \mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g} \longrightarrow \underline{\mathbb{Z}}_{\mathcal{M}_g} \longrightarrow 0\ ,$$

from which we obtain the long exact sequence

$$\mathrm{H}^0(\mathcal{M}_g,\underline{\mathbb{Z}}_{\mathcal{M}_g}) \xrightarrow{\delta} \mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \longrightarrow \mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \longrightarrow \mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\underline{\mathbb{Z}}_{\mathcal{M}_g}) = 0.$$

Here the vanishing of $\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\underline{\mathbb{Z}}_{\mathcal{M}_g})$ is due to [CTS21, Proposition 2.4.2]. Keeping track of the representing cocycles, the connecting homomorphism δ sends the element $1 \in \mathrm{H}^0(\mathcal{M}_g,\underline{\mathbb{Z}}_{\mathcal{M}_g})$ to the class of the torsor $[\mathrm{Pic}^1_{\mathcal{M}_{g,1}/\mathcal{M}_g}]$.

By Proposition 1, there are rational points on $\mathrm{Pic}^d_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ if and only if d is a multiple of $2g-2$, hence the period of $[\mathrm{Pic}^1_{\mathcal{M}_{g,1}/\mathcal{M}_g}]$ equals $2g-2$. Therefore, showing that

$$\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \cong \mathbb{Z}/(2g-2)\mathbb{Z}\cdot[\mathrm{Pic}^1_{\mathcal{M}_{g,1}/\mathcal{M}_g}],$$

is equivalent to showing

$$\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g,\mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g}) = 0.$$

3.2. **Reduction to injectivity.** We consider the Leray spectral sequence of $\mathbb{G}_m$ along the projection $\pi\colon \mathcal{M}_{g,1} \to \mathcal{M}_g$. Since π is a family of smooth curves over a smooth base, we have the following lemma:

Lemma 1. *We have* $\mathrm{R}^i\pi_*\mathbb{G}_m = 0$ *for all* $i \geq 2$.

Proof. By the Kummer sequence on $\mathcal{M}_{g,1}$, for any n we have the following short exact sequence of direct image sheaves on $\mathcal{M}_g$

$$0 \longrightarrow \mathrm{R}^i\pi_*\mathbb{G}_m \otimes \mathbb{Z}/n\mathbb{Z} \xrightarrow{\delta} \mathrm{R}^{i+1}\pi_*\mu_n \longrightarrow \mathrm{R}^{i+1}\pi_*\mathbb{G}_m[n] \longrightarrow 0\ .$$

Since $\mathcal{M}_{g,1}$ is regular noetherian, by [CTS21, Lemma 3.5.3] we know that for $i \geq 1$, the sheaves $\mathrm{R}^{i+1}\pi_*\mathbb{G}_m$ are torsion, therefore it suffices to show $\mathrm{R}^{i+1}\pi_*\mathbb{G}_m[n] = 0$ for all n.

For $i = 1$, we know $\mathrm{R}^1\pi_*\mathbb{G}_m$ is the relative Picard sheaf, and the connecting homomorphism δ is the degree map. Since one can always find relative degree-1 line bundles after étale localization, the degree map is stalkwise surjective, so $\mathrm{R}^2\pi_*\mathbb{G}_m[n] = 0$. For $i \geq 2$, we know $\mathrm{R}^{i+1}\pi_*\mu_n = 0$ by proper base change, hence $\mathrm{R}^{i+1}\pi_*\mathbb{G}_m[n] = 0$. □

With this lemma, we see that the Leray spectral sequence

$$\mathrm{H}^p_{\text{ét}}(\mathcal{M}_g, \mathrm{R}^q\pi_*\mathbb{G}_m) \Rightarrow \mathrm{H}^{p+q}_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m)$$

is concentrated in two rows, and gives rise to the following long exact sequence

$$\mathrm{H}^2_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m) \xrightarrow{\pi^*} \mathrm{H}^2_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m) \longrightarrow \mathrm{H}^1_{\text{ét}}(\mathcal{M}_g, \mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g})$$

$$\xrightarrow{\mathrm{d}_2^{1,1}} \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m) \xrightarrow{\pi^*} \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m)$$

By Proposition 2, the vanishing of $\mathrm{H}^1_{\text{ét}}(\mathcal{M}_g, \mathrm{Pic}_{\mathcal{M}_{g,1}/\mathcal{M}_g})$ is equivalent to the injectivity of

$$\pi^*\colon \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m) \to \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m)$$

for $g \geq 3$. By [CTS21, Lemma 3.5.3], the higher étale cohomology groups are torsion, therefore it suffices to show the injectivity on ℓ^∞-torsion part for all primes ℓ:

$$\pi^*\colon \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m)[\ell^\infty] \to \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m)[\ell^\infty].$$

3.3. **Applying the stability result.** By taking the direct limit of the Kummer sequences, we have a commutative diagram of short exact sequences

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathrm{H}^2_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell & \longrightarrow & \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mu_{\ell^\infty}) & \longrightarrow & \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mathbb{G}_m)[\ell^\infty] & \longrightarrow & 0\ . \\
 & & \downarrow{\scriptstyle \pi^*\otimes\mathbb{Q}_\ell/\mathbb{Z}_\ell} & & \downarrow{\scriptstyle \pi^*} & & \downarrow{\scriptstyle \pi^*} & & \\
0 & \longrightarrow & \mathrm{H}^2_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m) \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell & \longrightarrow & \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mu_{\ell^\infty}) & \longrightarrow & \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mathbb{G}_m)[\ell^\infty] & \longrightarrow & 0
\end{array}$$

Applying the snake lemma to the above diagram yields a short exact sequence

$$\ker(\pi^*|_{\mathrm{H}^3(\mathcal{M}_g, \mu_{\ell^\infty})}) \longrightarrow \ker(\pi^*|_{\mathrm{H}^3(\mathcal{M}_g, \mathbb{G}_m)[\ell^\infty]}) \longrightarrow \mathrm{coker}(\pi^*|_{\mathrm{H}^2(\mathcal{M}_g, \mathbb{G}_m)} \otimes \mathbb{Q}_\ell/\mathbb{Z}_\ell)\ .$$

The third term vanishes by Proposition 2, therefore we reduce our problem to showing the injectivity of

$$\pi^*\colon \mathrm{H}^3_{\text{ét}}(\mathcal{M}_g, \mu_{\ell^\infty}) \to \mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,1}, \mu_{\ell^\infty}).$$

By the comparison theorem for the étale and complex topologies [Mil80, Theorem 3.12], after fixing roots of unity, we may identify $\mathrm{H}^3_{\text{ét}}(\mathcal{M}_{g,n}, \mu_{\ell^\infty})$ with $\mathrm{H}^3(\mathcal{M}^{\text{an}}_{g,n}, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$. This further reduces Theorem 1 to proving the injectivity of the pullback map on singular cohomology

$$\pi^*\colon \mathrm{H}^3(\mathcal{M}^{\text{an}}_g, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \to \mathrm{H}^3(\mathcal{M}^{\text{an}}_{g,1}, \mathbb{Q}_\ell/\mathbb{Z}_\ell).$$

Since $\mathbb{Q}_\ell/\mathbb{Z}_\ell$ is an injective module, by the universal coefficient theorem, for any topological space X, we have natural isomorphism

$$\mathrm{H}^3(X, \mathbb{Q}_\ell/\mathbb{Z}_\ell) \cong \mathrm{Hom}(\mathrm{H}_3(X, \mathbb{Z}), \mathbb{Q}_\ell/\mathbb{Z}_\ell).$$

Thus the injectivity of π^* will be implied by the surjectivity of

$$\pi_*\colon \mathrm{H}_3(\mathcal{M}^{\text{an}}_{g,1}, \mathbb{Z}) \to \mathrm{H}_3(\mathcal{M}^{\text{an}}_g, \mathbb{Z}).$$

By the discussion in Section 2.2, this is equivalent to the surjectivity of

$$\mathrm{H}_3(\epsilon_g)\colon \mathrm{H}_3(\Gamma^1_g, \mathbb{Z}) \to \mathrm{H}_3(\Gamma_g, \mathbb{Z}),$$

which follows from Corollary 1.

4. Finishing the proof of Proposition 2

4.1. **Some topological results.** For finite locally constant sheaves, we identify étale cohomology with complex cohomology. Let us fix a smooth genus-g curve C and represent the local system $\mathrm{R}^i\pi_*\mu_m$ on $\mathcal{M}_g$ via the Γ_g action on $\mathrm{H}^i(C, \mu_m)$. Poincaré duality naturally identifies $\mathrm{H}^1(C, \mathbb{Z})$ with $\mathrm{H}_1(C, \mathbb{Z})$, we denote both by $\mathbb{H}$.

As remarked in [HR01, Corollary 22], the base and fiber of $\mathrm{Pic}^d_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ are Eilenberg-MacLane spaces, so is the total space, giving a short exact sequence of fundamental groups

$$0 \longrightarrow \mathbb{H} \longrightarrow \pi_1(\mathrm{Pic}^d_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \longrightarrow \Gamma_g \longrightarrow 1\ ,$$

which corresponds to an extension class $\epsilon(d)$ in $\mathrm{H}^2(\Gamma_g, \mathbb{H})$.

Lemma 2. *Under the identification* $\mathrm{H}^2(\Gamma_g, \mathbb{H}) \cong \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z})$*, up to a* $\pm$ *sign which depends on convention, this class satisfies* $\epsilon(1) = \mathrm{d}_{2,\mathbb{Z}}^{0,1}(1)$*, where* $\mathrm{d}_{2,\mathbb{Z}}^{0,1}\colon \mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*\mathbb{Z}) \to \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z})$ *is the transgression map.*

Proof. The transgression map can be calculated by Čech cohomology with respect to the Teichmüller covering $\{\mathcal{T}_g \to \mathcal{M}_g\}$. The Čech nerve $T_g^{\times_{\mathcal{M}_g} k+1} \cong (\Gamma_g)^k \times \mathcal{T}_g$ and the Čech complex is isomorphic to the bar complex for group cohomology of Γ_g. □

Lemma 3. [HR01, Theorem 24] *For* $g \geq 3$*, the map* $\mathbb{Z}/(2g-2)\mathbb{Z} \to \mathrm{H}^2(\Gamma_g, \mathbb{H})$ *that takes* $\overline{d}$ *to the class* $\epsilon(d)$ *corresponding to* $\mathrm{Pic}^d_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ *is a group isomorphism.*

4.2. **The key statement.** In Section 2.3, Proposition 2 was reduced to the statement that

$$\mathrm{H}^2(\mathcal{M}_{g,1}, \mu_m) = \pi^*\mathrm{H}^2(\mathcal{M}_g, \mu_m) + \delta(\mathrm{Pic}(\mathcal{M}_{g,1})).$$

We prove, independent of [FP21, Theorem 4.1], that the statement holds for all $g \geq 3$.

Proposition 4. *For the Leray spectral sequence of* μ_m *along* $\pi\colon \mathcal{M}_{g,1} \to \mathcal{M}_g$*, we have:*

(1) $E_r^{0,2}$ *stabilizes on the* E_3 *page, and equals the subgroup of* $\mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*\mu_m)$ *generated by the class of the relative canonical bundle* $\mathcal{L} := \omega_{\mathcal{M}_{g,1}} \otimes \pi^*\omega_{\mathcal{M}_g}^{\otimes -1}$*.*

(2) $E_r^{1,1}$ *stabilizes on the* E_2 *page, and* $\mathrm{H}^1(\mathcal{M}_g, \mathrm{R}^1\pi_*\mu_m) \cong \mathbb{Z}/(m, 2g-2)\mathbb{Z}$*. Furthermore, the group is generated by the image of* $\mathcal{L}^{\otimes \frac{m}{(m,2g-2)}}$ *in* $F^1/F^2 = E_\infty^{1,1}$*.*

Therefore any class in $\mathrm{H}^2(\mathcal{M}_{g,1}, \mu_m)$ *can be modified to* $\pi^*\mathrm{H}^2(\mathcal{M}_g, \mu_m)$ *by adding suitable multiple of* $\mathcal{L}$*, hence* $\mathrm{H}^2(\mathcal{M}_{g,1}, \mu_m) = \pi^*\mathrm{H}^2(\mathcal{M}_g, \mu_m) + \delta(\mathrm{Pic}(\mathcal{M}_{g,1}))$*.*

Proof. (1) In order to determine $E_r^{0,2}$, we first look at the transgression map

$$\mathrm{d}_2^{0,2}\colon \mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*\mu_m) \to \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mu_m),$$

In general the map is given by taking Yoneda product with the class $c \in \mathrm{Ext}^2(\mathrm{R}^2\pi_*\mu_m, \mathrm{R}^1\pi_*\mu_m)$ given via the triangle

$$\mathrm{R}^1\pi_*\mu_m \longrightarrow \tau_{[1,2]}\mathrm{R}\pi_*\mu_m \longrightarrow \mathrm{R}^2\pi_*\mu_m[-1] \xrightarrow{c} \mathrm{R}^1\pi_*\mu_m[1]\ .$$

Since Kummer sequence shows that μ_m is quasi-isomorphic to the complex $[\ \mathbb{G}_m \xrightarrow{m} \mathbb{G}_m\ ]$ concentrated in degrees $0, 1$, using Lemma 1, we see that the transgression map can be

obtained by taking cup product with the extension class of

$$0 \longrightarrow \mathrm{R}^1\pi_*\mu_m \longrightarrow \mathrm{R}^1\pi_*\mathbb{G}_m \xrightarrow{m} \mathrm{R}^1\pi_*\mathbb{G}_m \longrightarrow \mathrm{R}^2\pi_*\mu_m \longrightarrow 0 \ .$$

We wish to identify this map with the topological description in Lemma 3. Choose a primitive m-th root of unity ζ_m and consider the commutative diagram of analytic sheaves on $\mathcal{M}_{g,1}$

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathcal{O} & \xrightarrow{\exp} & \mathcal{O}^* & \longrightarrow & 0 \\ & & \downarrow & & \downarrow{\scriptstyle \exp} & & \downarrow{\scriptstyle m} & & \\ 0 & \longrightarrow & \mathbb{Z}/m\mathbb{Z}\cdot\zeta_m & \longrightarrow & \mathcal{O}^* & \xrightarrow{m} & \mathcal{O}^* & \longrightarrow & 0 \end{array}$$

it yields the commutative diagram of direct images

$$\begin{array}{ccccccc} \mathrm{R}^1\pi_*\mathbb{Z} & \longrightarrow & \mathrm{R}^1\pi_*\mathcal{O} & \xrightarrow{\exp} & \mathrm{R}^1\pi_*\mathcal{O}^* & \xrightarrow{\delta} & \mathrm{R}^2\pi_*\mathbb{Z} \\ \downarrow & & \downarrow{\scriptstyle \exp} & & \downarrow{\scriptstyle m} & & \downarrow \\ \mathrm{R}^1\pi_*\mathbb{Z}/m\mathbb{Z} & \longrightarrow & \mathrm{R}^1\pi_*\mathcal{O}^* & \xrightarrow{m} & \mathrm{R}^1\pi_*\mathcal{O}^* & \xrightarrow{\delta} & \mathrm{R}^2\pi_*\mathbb{Z}/m\mathbb{Z}. \end{array}$$

Naturality of the Yoneda pairing gives us commutative diagram

$$\begin{array}{ccccc} \mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*\mathbb{Z}) & \xrightarrow{\mathrm{d}_{2,\mathbb{Z}}^{0,2}} & \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z}) & = & \mathrm{H}^1(\Gamma_g, \mathbb{H}) \cong \frac{\mathbb{Z}}{(2g-2)\mathbb{Z}}\cdot\epsilon(1) \\ \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \\ \mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*(\mathbb{Z}/m\mathbb{Z})) & \xrightarrow{\mathrm{d}_2^{0,2}} & \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z})) & = & \mathrm{H}^2(\Gamma_g, \mathbb{H}/m\mathbb{H}) \end{array}$$

the left vertical map α is quotient by m. Running the Bockstein sequence

$$0 \longrightarrow \mathrm{R}^1\pi_*\mathbb{Z} \xrightarrow{m} \mathrm{R}^1\pi_*\mathbb{Z} \longrightarrow \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z}) \longrightarrow 0$$

we see that β factors through the natural injection

$$i\colon \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z}) \otimes (\mathbb{Z}/m\mathbb{Z}) \to \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z})).$$

By Lemma 3 and Lemma 2, under the identification

$$\mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z}) \otimes (\mathbb{Z}/m\mathbb{Z}) \cong \mathrm{H}^2(\Gamma_g, \mathbb{H}) \otimes (\mathbb{Z}/m\mathbb{Z}) \cong \mathbb{Z}/(m, 2g-2)\mathbb{Z}\cdot\epsilon(1),$$

the order of $\mathrm{d}_2^{0,2}(\overline{1})$ equals $(m, 2g-2)$, hence $\ker(\mathrm{d}_2^{0,2})$ is the subgroup of $\mathrm{H}^0(\mathcal{M}_g, \mathrm{R}^2\pi_*\mu_m) = \mathbb{Z}/m\mathbb{Z}$ generated by the class $\overline{2g-2}$. This class is represented by the relative canonical bundle, and survives to the E_∞ page, therefore $E_{r\geq 3}^{0,2}$ stabilizes.

(2) Since $\deg \mathcal{L} = 2g-2$, the degree of $\mathcal{L}^{\otimes \frac{m}{(m,2g-2)}}$ is divisible by m, so its chern class $\eta = c_1(\mathcal{L}^{\otimes \frac{m}{(m,2g-2)}})$ lies in F^1. We aim to show that the image $\overline{\eta}$ generates $F^1/F^2 \subseteq \mathrm{H}^1(\mathcal{M}_g, \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z}))$.

As argued in [HR01, Proposition 23], for $g \geq 2$, we have $\mathrm{H}^1(\Gamma_g, \mathbb{H}) = 0$. Therefore the boundary map induced by the Bockstein sequence gives an isomorphism

$$\delta \colon \mathrm{H}^1(\mathcal{M}_g, \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z})) \cong \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z})[m].$$

By Lemma 3, the group $\mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z})[m] = \mathrm{H}^2(\Gamma_g, \mathbb{H})[m]$ is generated by $\frac{2g-2}{(m,2g-2)} \cdot \epsilon(1)$, where $\epsilon(1) = \mathrm{d}_2^{0,1}(1)$ up to a sign. Since η is the cohomology class of a global line bundle, it survives to the E_∞ page. Therefore it suffices to show that

$$\delta(\eta) = \frac{2g-2}{(m, 2g-2)} \cdot \mathrm{d}_{2,\mathbb{Z}}^{0,1}(1).$$

- First we calculate $\eta = c_1(\mathcal{L}^{\otimes \frac{2g-2}{(2g-2,m)}})$, using that the chern class is the connecting homomorphism of the Kummer sequence. We pick an m-th root line bundle $\mathcal{N}_i \in \pi^{-1}(U_i)$ for $\mathcal{L}^{\otimes \frac{m}{(m,2g-2)}}$ on some open covering $\{U_i\}_{i\in I}$. This yields a 0-cocycle. Since $\mathcal{N}_i^{\otimes m}|_{U_{ij}} \cong \mathcal{L}|_{U_{ij}} \cong \mathcal{N}_j^{\otimes m}|_{U_{ij}}$, after taking coboundary, the Čech 1-cocycle takes values in $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}[m]$.
- We identify $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}[m]$ with $\mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z})$ using the exact sequence of topological uniformization

$$0 \longrightarrow \mathrm{R}^1\pi_*\mathbb{Z} \longrightarrow \mathrm{R}^1\pi_*\mathbb{R} \cong R^1\pi_*\mathcal{O} \longrightarrow \mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g} \longrightarrow 0 \ ,$$

  where the identification in the middle is obtained via Hodge theory using inclusion and projection $\mathrm{R}^1\pi_*\mathbb{R} \to \mathrm{R}^1\pi_*\mathbb{C} \to \mathrm{R}^1\pi_*\mathcal{O}$. For any local section h_i of $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}[m]$, we lift it to $\widetilde{h}_i$ on $\mathrm{R}^1\pi_*\mathbb{R}$. Since $mh_i = 0$, we know that $m\widetilde{h}_i$ is a local section of $\mathrm{R}^1\pi_*\mathbb{Z}$. Note that for different choices of h_i, the section $m\widetilde{h}_i$ differ by $m\mathrm{R}^1\pi_*\mathbb{Z}$, therefore the identification $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g} \to \mathrm{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z}),\ h_i \mapsto m\widetilde{h}_i$ is well-defined. The class η is represented by the cocycle which takes value $m\widetilde{n}_{ij}$ on U_{ij}, where $\widetilde{n}_{ij}$ are local sections to $\mathrm{R}^1\pi_*\mathbb{R}$ that represents $\mathcal{N}_i \otimes \mathcal{N}_j^{\otimes -1}$.
- The class $\delta(\eta)$ is calculated by tracing back the Bockstein sequence. Lifting the local sections $m\widetilde{n}_{ij}$ in $\mathbb{R}^1\pi_*(\mathbb{Z}/m\mathbb{Z})$ to $m\widetilde{n}_{ij}$ in $\mathbb{R}^1\pi_*\mathbb{Z}$, taking boundary then divide by m, we get the 2-coboundary that takes value $\widetilde{n}_{ij} - \widetilde{n}_{jk} + \widetilde{n}_{ki}$ on U_{ijk}.

- On the other hand, using the previous uniformization sequence, the class of the torsor $\mathrm{Pic}^{\frac{2g-2}{(m,2g-2)}}_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ is obtained by the connecting homomorphism $\mathrm{H}^1(\mathcal{M}_g, \mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}) \to \mathrm{H}^2(\mathcal{M}_g, \mathrm{R}^1\pi_*\mathbb{Z})$ which acts on the 1-cocycle $\mathcal{N}_i \otimes \mathcal{N}_j^{\otimes -1}$ by lifting the section in $\mathrm{Pic}^0_{\mathcal{M}_{g,1}/\mathcal{M}_g}$ to $\widetilde{n}_{ij}$ in $\mathrm{R}^1\pi_*\mathbb{R}$ then taking coboundary. This coincides with the class of $\delta(\eta)$. This 2-cocycle represents the class of the Picard torsor $\mathrm{Pic}^{\frac{2g-2}{(m,2g-2)}}_{\mathcal{M}_{g,1}/\mathcal{M}_g}$, since $\mathcal{N}_i$ are line bundles of degree $\frac{2g-2}{(m,2g-2)}$.

Therefore we have proved that F^1/F^2 is generated by $\mathcal{L}^{\otimes \frac{2g-2}{(m,2g-2)}}$, hence $\mathrm{H}^2(\mathcal{M}_g, \mu_m) = \pi^*\mathrm{H}^2(\mathcal{M}_g, \mu_m) + \delta(\mathrm{Pic}(\mathcal{M}_{g,1}))$ for all $g \geq 3$. □

4.3. **Relation with** [KS03, Theorem 1.2]**.** By Proposition 2, we calculate that $\mathrm{Br}'(\mathcal{M}_{3,1}) = \mathbb{Z}/2\mathbb{Z}$ over $\mathbb{C}$. By [FP21, Proposition 4.1], we have

$$\mathrm{Br}'(\mathcal{M}_{3,1}) = \mathrm{H}_2(\mathcal{M}_{3,1}, \mathbb{Z})_{\mathrm{tors}} = \mathrm{H}_2(\Gamma^1_3, \mathbb{Z})_{\mathrm{tors}}.$$

In [KS03, Theorem 1.2, Proposition 1.4], it is proved that

$$\mathrm{H}_2(\Gamma_3, \mathbb{Z}) = A_1 \oplus \mathbb{Z},\ \mathrm{H}_2(\Gamma_{3,1}, \mathbb{Z}) = A_2 \oplus \mathbb{Z},$$

$$\mathrm{H}_2(\Gamma^1_3, \mathbb{Z}) = \mathrm{H}_2(\Gamma_{3,1}, \mathbb{Z}) \oplus \mathbb{Z} = A_2 \oplus \mathbb{Z}^{\oplus 2},$$

with $0 \subseteq A_2 \subseteq A_1 \subseteq \mathbb{Z}/2\mathbb{Z}$. Therefore $A_1 = A_2 = \mathbb{Z}/2\mathbb{Z}$ and

(1) $\mathrm{H}_2(\Gamma_{3,1}, \mathbb{Z}) = (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}$, $\mathrm{H}_2(\Gamma_3, \mathbb{Z}) = (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}$.
(2) $\mathrm{H}_2(\Gamma^1_3, \mathbb{Z}) = (\mathbb{Z}/2\mathbb{Z}) \oplus \mathbb{Z}^{\oplus 2}$.

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